

Objectives

This phase focuses on the procurement, cleaning, and processing of a large-scale dataset consisting of at least **500,000 student records** for use in the project. Data are collected from appropriate and ethical sources in accordance with established data privacy and ethical standards. The acquired dataset undergoes cleaning and preprocessing procedures, including the treatment of missing values, removal of duplicate records, standardization of data formats, and preparation of relevant academic and psychological indicators. Throughout the process, data anonymization protocols and ethical guidelines are strictly observed to ensure the responsible handling, storage, and utilization of information within the big data pipeline.

Tasks Completed

- **Large-Scale Data Procurement:** Collected and consolidated a student dataset exceeding the minimum requirement of 500,000 records to support large-scale analytics and model development.
- **Data Cleaning and Validation:** Performed preprocessing activities such as handling missing values, removing duplicate entries, standardizing formats, and verifying the completeness of academic and psychological indicators.
- **Ethical Data Preparation and Ingestion:** Ensured compliance with data privacy and ethical guidelines through data anonymization procedures before loading the processed dataset into the PySpark environment for distributed processing and analysis.

Tasks to be Completed

N/A

Results and Analysis

- Successfully acquired and validated 1,000,000 distinct student records, exceeding the project's minimum requirement of 500,000 records. The dataset contained essential academic and psychological indicators, including burnout, anxiety, and sleep hours, supporting the project's analytical objectives.
- Data cleaning and preprocessing procedures effectively resolved missing values, duplicate entries, and formatting inconsistencies, resulting in a high-quality and reliable dataset suitable for large-scale analytics and machine learning applications.
- Compliance checks confirmed that all records were fully anonymized and adhered to data privacy standards. The processed dataset was successfully ingested into the PySpark environment,

demonstrating the capability of the big data pipeline to efficiently handle large-scale data processing.

Challenges and Solutions

During development, the code occasionally encountered errors due to missing libraries and dependencies. To resolve this issue, regular checks were conducted to verify that all required libraries were installed and updated, ensuring compatibility and the successful execution of the program.

Conclusion

The data gathering and preprocessing phase was successfully completed through the acquisition of 1,000,000 student records, surpassing the project's minimum requirement of 500,000 records. Comprehensive data cleaning and validation procedures were performed to address missing values, duplicate entries, and formatting inconsistencies, resulting in a high-quality dataset suitable for large-scale analytics and machine learning applications. Ethical data handling practices, including data anonymization and privacy compliance checks, were strictly implemented before the dataset was ingested into the PySpark environment for distributed processing. Validation confirmed the presence of key academic and psychological indicators, such as burnout, anxiety, and sleep hours, which support the project's analytical objectives. A challenge encountered during this phase involved code execution errors caused by missing libraries and dependencies. This issue was addressed through regular verification and updating of required libraries, ensuring system compatibility and the successful execution of the data processing pipeline.